



## Real-world issue: Sensor faults

- Sometimes systems for which we would like a state estimate use sensors having intermittent faults.
- We would like to detect faulty measurements and discard them.
  - Time update steps of the KF still implemented.
  - Measurement update steps are skipped ( $L_k = 0$ ).
- KF provides elegant theoretical means to accomplish this goal. Background:
  - Predicted measurement is  $\hat{z}_k = C_k \hat{x}_k^- + D_k u_k$ .
  - Prediction covariance (uncertainty) matrix is  $\Sigma_{\tilde{z},k} = C_k \Sigma_{\tilde{x},k} C_k^T + \Sigma_{\tilde{v}}$ .
  - The innovation is  $\tilde{z}_k = z_k - \hat{z}_k$ .
- By combining  $\tilde{z}_k$  and  $\Sigma_{\tilde{z},k}$  we can determine if the innovation is “too big,” which indicates a possible sensor fault.



## Measurement validation gating

- Can place “measurement validation gate” on measurement using normalized estimation error squared (NEES):

$$e_k^2 = \tilde{z}_k^T \Sigma_{\tilde{z},k}^{-1} \tilde{z}_k.$$

- NEES  $e_k^2$  has Chi-squared distribution with  $m$  degrees of freedom, where  $z_k \in \mathbb{R}^m$ .
- If  $e_k^2$  is outside of bounding value for Chi-squared distribution for a desired confidence level, then measurement is discarded.
- Note: If a many measurements are discarded in a short time interval, the sensor may truly have failed, or the state estimate and covariance may have gotten “lost.”
- It is sometimes helpful to “bump up” covariance  $\Sigma_{\tilde{x},k}^{\pm}$ , which simulates additional process noise, to help Kalman filter to reacquire.
- Both done in practice to aid robustness of a real implementation.



## NEES is chi-squared

- To prove NEES is chi-squared, define  $y_k = M_k \tilde{z}_k$ .
  - Mean of  $y_k$  is  $\mathbb{E}[y_k] = \mathbb{E}[M_k \tilde{z}_k] = 0$ .
  - Covariance of  $y_k$  is  $\Sigma_{\tilde{y},k} = \mathbb{E}[M_k \tilde{z}_k \tilde{z}_k^T M_k^T] = M_k \Sigma_{\tilde{z},k} M_k^T$ .
  - $y_k$  is Gaussian (since it is a linear combination of Gaussians).
- If we define  $M_k$  such that  $M_k^T M_k = \Sigma_{\tilde{z},k}^{-1}$ , then:
  - $M_k$  is the lower-triangular Cholesky factor of  $\Sigma_{\tilde{z},k}^{-1}$ .
  - Also,  $y_k \sim \mathcal{N}(0, I)$  since:

$$\begin{aligned} \Sigma_{\tilde{y},k} &= M_k (M_k^T M_k)^{-1} M_k^T \\ &= M_k M_k^{-1} M_k^{-T} M_k^T = I. \end{aligned}$$

- NEES  $e_k^2 = y_k^T y_k = \tilde{z}_k^T \Sigma_{\tilde{z},k}^{-1} \tilde{z}_k$  is the sum of squares of independent  $\mathcal{N}(0, 1)$  RVs.
- So,  $e_k^2$  is chi-square with  $m$  degrees of freedom, where  $m$  is the dimension of  $\tilde{z}_k$ .

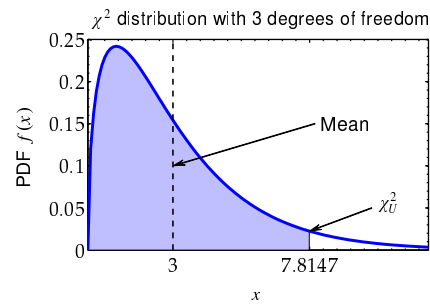


## What does this really mean?

- $e_k^2$  never negative (sum of squares); pdf also asymmetric.
- pdf of chi-square RV  $X$  having  $m$  degrees of freedom is:

$$f_X(x) = \frac{1}{2^{m/2} \Gamma(m/2)} x^{(m/2-1)} e^{-m/2}.$$

- Tricky, but don't need to evaluate in real time.
- Instead, use value *precomputed* from pdf.
- For  $1 - \alpha$  confidence of a valid measurement, need to find  $\chi_U^2$  such that there is  $\alpha$  area above  $\chi_U^2$  (figure drawn for  $\alpha = 0.05$ ).
- Discard measurement if NEES greater than  $\chi_U^2$ .



## Computer calculation of $\chi_U^2$

- In MATLAB (Statistics and Machine Learning Toolbox) can find  $\chi_U^2$  where inverse CDF is equal to  $1 - \alpha$ :  
`X2U = chi2inv(1-0.01,2) % Upper critical value X2U = 9.2103`
- Function "chi2inv" is built in to Octave.
- Note that  $\chi_U^2$  needs to be computed once only, offline.
  - Based only on  $m$  and  $\alpha$ , so doesn't need to be recalculated as KF runs.
- For hand calculations a  $\chi^2$ -table is available on next page.
- If  $e_k^2 > \chi_U^2$ , then measurement is discarded ( $L_k = 0$ ); else, measurement kept.



## Manual table-lookup of $\chi_U^2$

- For chi-squared distribution with  $m$  degrees of freedom, table entries list values of  $\chi_U^2(\alpha, m)$  for specified upper tail area  $\alpha$ :

| Degrees of freedom $m$ | Upper tail areas $\alpha$ |        |        |        |        |        |
|------------------------|---------------------------|--------|--------|--------|--------|--------|
|                        | 0.25                      | 0.10   | 0.05   | 0.025  | 0.01   | 0.005  |
| 1                      | 1.323                     | 2.706  | 3.841  | 5.024  | 6.635  | 7.879  |
| 2                      | 2.773                     | 4.605  | 5.991  | 7.378  | 9.210  | 10.597 |
| 3                      | 4.108                     | 6.251  | 7.815  | 9.348  | 11.345 | 12.838 |
| 4                      | 5.385                     | 7.779  | 9.488  | 11.143 | 13.277 | 14.860 |
| 5                      | 6.626                     | 9.236  | 11.070 | 12.833 | 15.086 | 16.750 |
| 6                      | 7.841                     | 10.645 | 12.592 | 14.449 | 16.812 | 18.548 |



## Integration into the Kalman filter

```
% KF Step 1c: Predict system output
zhat = Cd*xhat + Dd*u(:,k);
zerror = z(:,k) - zhat;
SigmaZ = C*SigmaX*C' + SigmaV;
nees = zerror'/SigmaZ*zerror;

% KF validation gate (X2U can be calculated outside of loop)
alpha = 0.01; confidence = 1 - alpha;
X2U = chi2inv(confidence,length(zhat)); % Upper critical value

if nees <= X2U
    % KF Step 2a: Compute Kalman gain matrix
    L = SigmaX*C'/SigmaZ;

    % KF Step 2b: State estimate measurement update
    xhat = xhat + L*zerror;

    % KF Step 2c: Estimation-error covariance measurement update
    SigmaX = SigmaX - L*C*SigmaX;
end
```



## Summary

- KF has built-in mechanism that enables detecting sensor errors.
- Once only, off-line, precompute  $\chi_U^2(\alpha, m)$  for  $z_k \in \mathbb{R}^m$  and desired  $\alpha$ .
- As KF executes, every time sample, compute  $e_k^2 = \tilde{z}_k^T \Sigma_{\tilde{z},k}^{-1} \tilde{z}_k$ .
  - If  $e_k^2 > \chi_U^2(\alpha, m)$ , then discard measurement (set  $L_k = 0$ ).
  - Otherwise, apply measurement update as usual.
- If many sequential measurements discarded, consider “bumping up” covariance as  $\Sigma_{\tilde{x},k}^+ = Q \Sigma_{\tilde{x},k}^+$  where  $Q > 1$ .
- If problems persist, likely a permanent sensor fault.